



**AGRICORE: A GAMIFIED WEB-BASED LEARNING PLATFORM FOR PRINCIPLES
OF CROP PROTECTION I FOR BS AGRICULTURE
AT DAVAO ORIENTAL STATE UNIVERSITY**

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CHAPTER I

INTRODUCTION

1.1 Rationale of the Study

Principles of Crop Protection I is an important course in the BS Agriculture program that develops students' skills in identifying, diagnosing, and managing crop pests, diseases, and weeds. These skills require students to understand technical concepts and apply them to practical agricultural situations. Since students have different learning paces and learning needs, maintaining engagement and providing opportunities for repeated practice can be challenging in conventional learning settings (Murniati et al., 2023; Tikhonova et al., 2022).

At Davao Oriental State University (DOrSU), students have limited opportunities to repeatedly practice crop-protection diagnosis before their scheduled practical activities. Difficulties in identifying pests, diseases, and other crop problems may affect their performance during field activities. Students also need to study numerous scientific names, symptoms, and characteristics, which can make independent learning repetitive and difficult to sustain. Gamification and interactive learning activities can provide feedback, encourage participation, and allow students to practice in a low-risk environment (Deterding et al., 2011; Kapp, 2012).

Existing agricultural learning resources provide valuable information but are primarily text-based or designed for broader agricultural applications. There is a need for a digital learning platform that combines syllabus-aligned content with interactive and gamified crop-protection practice. To address this need, the study proposes AgriCore, a web-based learning system that follows a Learn-Practice-Apply approach. Through interactive modules, mission-based activities, XP, progress tracking, and leaderboards, AgriCore aims to provide students with additional opportunities to practice crop-protection

concepts and strengthen their preparation for practical field activities.

1.2 Purpose and Project Description

AgriCore will be developed as a gamified web-based learning system designed to support how BS Agriculture students at Davao Oriental State University (DOrSU) study *Principles of Crop Protection I*. The system will supplement classroom instruction by providing a unified digital platform that connects theoretical lessons with interactive practice. Through gamification elements such as Experience Points (XP), leaderboards, countdowns, and badges, AgriCore will make the study of pests, diseases, and other crop-protection topics more engaging and interactive.

The system will include modules for lesson delivery, interactive activities, and progress tracking, along with a student leaderboard and personalized dashboard for monitoring learning progress. It will also feature mission-based activities using real images of crop diseases and pests, allowing students to practice identification and management decisions in a risk-free environment. By providing structured digital practice and immediate feedback, AgriCore will support students in developing the diagnostic and decision-making skills needed for their laboratory activities and future fieldwork.

1.3 Objectives of the Study

General Objective

The goal of this project is to develop a web-based learning platform that offers e-learning combined with gamified features to help students assess and refine their understanding and skills in identifying crop threats and protection.

Specific Objectives:

1. To design and implement a content management and lesson delivery module that organizes learning materials according to the *Principles of Crop Protection I* syllabus.

2. To integrate gamification elements such as Experience Points (XP), leaderboards, countdowns, and badges to encourage student engagement and participation.
3. To develop interactive mission activities at the end of each module using different game mechanics to support the assessment of students' identification, diagnostic, and decision-making skills.
4. To implement a gamification and leaderboard system that tracks student XP, progress, and rankings within their assigned academic sections.
5. To evaluate the software quality of the platform using the ISO/IEC 25010 evaluation framework across the characteristics of functional suitability, performance efficiency, usability, and reliability.

User Evaluation Based on ISO/IEC 25010 Standards

To ensure that AgriCore meets international standards for software quality, the system will be evaluated using the ISO/IEC 25010 software quality model. This framework will assess the system as a software evaluation framework to ensure it delivers reliability, usability, security, and long-term value as a gamified learning platform for BS Agriculture students at Davao Oriental State University (DOrSU). The evaluation will focus on key characteristics such as functional suitability, performance efficiency, and maintainability to verify that the system effectively supports the technical requirements of the crop protection curriculum. By measuring these standard metrics, the researchers can ensure that the mission features and gamification elements are not only engaging but also stable and secure for students to use as a primary study tool outside of their prioritized practical hours.

1.4 Significance of the Study

The development of AgriCore addresses the problem of unorganized learning materials and passive learning approaches by integrating learning modules with

gamification features. This study is significant to the school community, specifically for BS Agriculture students, teachers, university administrators, and future researchers or developers. As we integrate normal learning modules with gamified features, the learning process becomes more exciting and effective for students.

Students. Students can benefit from the system as it eliminates the need to gather multiple unorganized learning sources. Furthermore, the system helps students to build and improve their skills in identifying crop threats and analyzing protection requirements in a fun and challenging way.

Teachers. Teachers will benefit from the learning modules and assessment features of AgriCore, which provide an interactive platform that they can recommend to their students. By encouraging students to use the platform, teachers can support their understanding of the subject and monitor their learning progress, helping identify students who may need additional guidance.

Future Developers. Future developers may use this study as a reference in building a new and more advanced gamified learning platform for the Principle of Crop Protection I.

1.5 Scope and Limitations of the Study

Scope

1. The system will focus on the digitization and enhancement of the Principles of Crop Protection I syllabus for undergraduate BS Agriculture students actively enrolled in the course at DOrSU - Main Campus, specifically transforming traditional learning processes into interactive, gamified learning environments.
2. The platform's content delivery boundary is strictly mapped to the five (5) core outline units of the official university syllabus: Module I (Introduction to Crop Protection), Module II (Plant Pathology), Module III (Agricultural Entomology),

Module IV (Weed Science), and Module V (Integrated Pest Management).

3. The AgriCore project will be developed as a web-based platform that is accessible for desktop computers, laptops and smartphones using any standard web browser.
4. The content delivery system supports diverse media formats, incorporating high-resolution photographs, short instructional video guides, and 2D animations to visually explain the course topics.
5. The system will reward student accuracy and module completion with Experience Points (XP), which serve as the data matrix for real-time leaderboard generation.
6. The system includes a teacher interface to manage student accounts, track overall progress, and see which topics students are mastering or struggling with.
7. The system will include five modules with different game types: Decision-Making, Identification, Matching, Drag-and-Drop, and Strategy and Management.

Syllabus Coverage

The AgriCore platform is based on the official Principles of Crop Protection I syllabus for Bachelor of Science in Agriculture students at Davao Oriental State University. The system covers five learning modules: Introduction to Crop Protection, Plant Pathology, Agricultural Entomology, Weed Science, and Integrated Pest Management. Each module includes selected syllabus topics and a corresponding game-based activity designed to support students' understanding and application of crop protection concepts. Table 1 presents the syllabus topics covered in each module and the corresponding game type implemented in AgriCore.

Table 1. Syllabus Coverage of AgriCore

Module	Covered Syllabus Topic	Game Type Description
Module 1: Introduction to Crop Protection	A. World Population and Food Supply, B. Role of Crop Protection,	Players read an agricultural crisis scenario presented by the guide character and must tap

	C. Definition of Pests D. Economic Importance of Pests E. Various Disciplines Involved in Crop Protection	and select one of three decision cards featuring specific farm actions, visual icons, and brief outcomes directly beneath the situation display to earn XP and advance through different module lessons.
Module 2: Plant Pathology	A. Plant Pathology B. Causes of Plant Diseases C. Disease Cycle D. Disease Epidemiology E. Variability of Plant Pathogens	Players use a zoom scope to inspect leaves for disease symptoms, marking analyzed areas with red circles to complete objectives. Once all spots are marked, identifying the correct pathogen profile below unlocks the NEXT LEVEL button to progress through Levels 1 to 8.
Module 3: Agricultural Entomology	A. Arthropod Pests B. Vertebrate Pests C. Natural Enemies of insects and vertebrate pests	Identify different arthropods and vertebrate pests based on their characteristics and recognize insect orders. Formulate appropriate management tactics.
Module 4: Weed Science	A. Weed Science B. Classification of Weeds C. Invasive Species	Players look at a weed photograph and must drag and drop structural labels (like specific root types, leaf shapes, or collars) directly onto flashing target zones on the plant picture to finish the task.
Module 5: Integrated Pest Management	A. Pest Problem Diagnosis B. Pest Management Concepts C. Pest Management Strategies D. Planning Pest Management Programs	A strategy and resource management game where players diagnose pest problems and select the best combination of eco-friendly tactics to save crops.

Limitations

1. The system depends on stable internet connectivity and access to digital devices; technical disruptions or low bandwidth may affect real-time data synchronization and leaderboard updates.
2. The system is delimited strictly to the curriculum of *Principles of Crop Protection I* and cannot be used for unrelated agricultural subjects or other degree programs without manual database and CMS adjustments.

1.6 Conceptual Framework

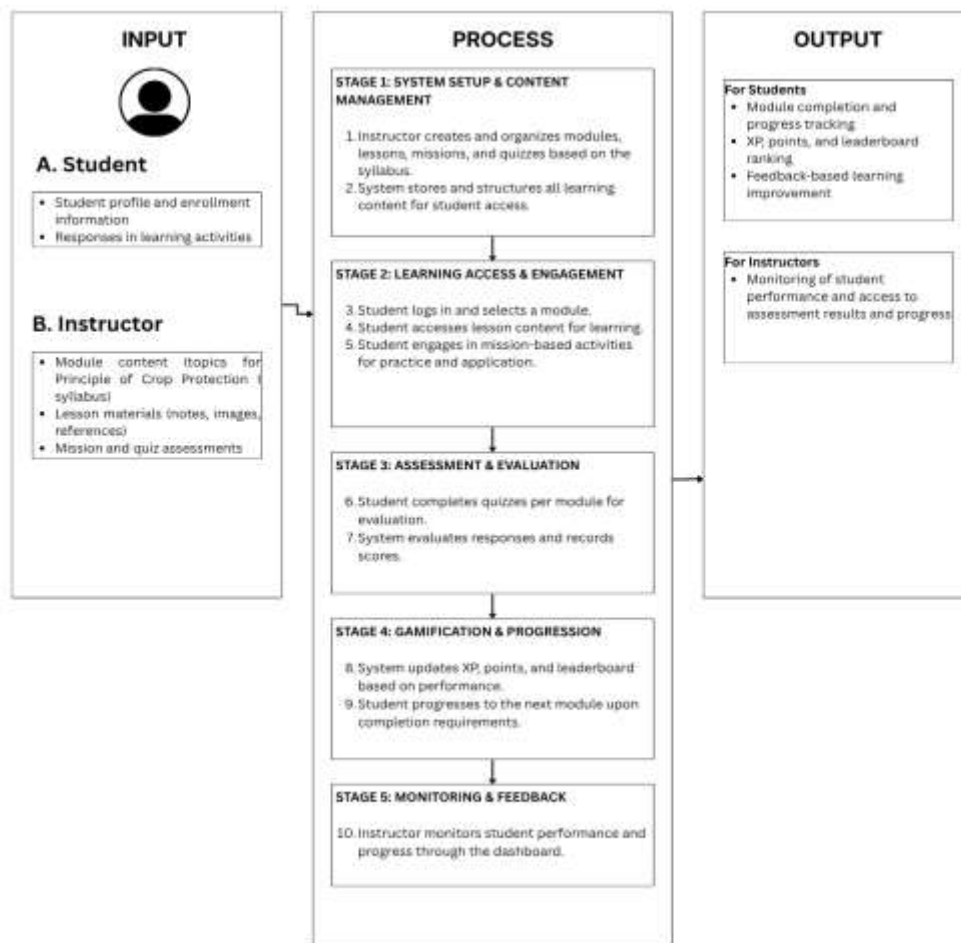


Figure 1. Conceptual Framework of AgriCore

The IPO model is used to explain the conceptual framework of AgriCore in figure 1. Data is provided by two users where instructors offer module content, lessons and quizzes for Crop Protection syllabus while students provide profile data and answers to the questions. System setting, learning access, assessment, gamification and monitoring are the five process stages where the data is offered, student answers are evaluated and monitored. Output will depend on the user where the instructors can monitor and evaluate the performances while students are offered progress tracking, XP and feedback. Taking all in consideration, the framework shows the ability of AgriCore to organize the input from users into a gamified learning process.

1.7 Definition of Terms

For the purpose of this study, the following terms are defined based on how they are used within the research and the system's operation:

Arthropod Pests

Represents the visual database of insect and non-insect invertebrate threats (e.g., mites) programmed into Module III, which serves as testing content to evaluate students' ability to classify feeding damage on host crops.

Badges

These are digital rewards given to students as proof of their achievements after completing a specific module or reaching certain milestones within the system.

Crop Protection

Pertains to the foundational subject domain encompassing weeds, pathogens, and animal pests, which provides the core learning objectives mapped directly into the platform's interactive assessments.

Entomology

Denotes Module III of the AgriCore platform, utilized in this research to test and measure student diagnostic accuracy regarding crop-damaging insects.

Experience Points (XP)

This refers to the cumulative points earned by students through completing missions and learning activities in the system. XP represents the student's overall learning progress and is used as a basis for leaderboard ranking.

Gamification

Signifies the integration of XP, badges, and leaderboards into the platform, serving as the core independent variable evaluated in this research to measure its effect on student participation.

Integrated Pest Management (IPM)	Consists of Module V of the AgriCore platform, structured as a strategy simulator where students adjust biological, chemical, and cultural variables to demonstrate yield-preservation competencies.
Leaderboard	Functions as the real-time ranking interface that aggregates student XP, used in this study to observe whether peer visibility promotes friendly competition and sustained platform usage.
Mission	A task-oriented simulation where students apply diagnostic knowledge to identify threats in a controlled digital environment.
Pathology (Plant Pathology)	Specifies Module II of the AgriCore platform, featuring symptom identification scenarios used to assess student mastery in recognizing fungal, bacterial, and viral crop diseases.
Principles of Crop Protection I	This refers to the specific technical course taken by BS Agriculture students at DORSU that serves as the basis for the system's content.

Vertebrate Pests

Applies to the interactive scenarios in Module III featuring crop-damaging backboned animals (such as rodents and birds), used to test student knowledge on field-level barrier management.

Weed Science

The study of unwanted plants that compete with crops for nutrients, water, and light. In the system, specifically within Module IV students will learn to identify different weed species and their impact on farm productivity.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Literature Map

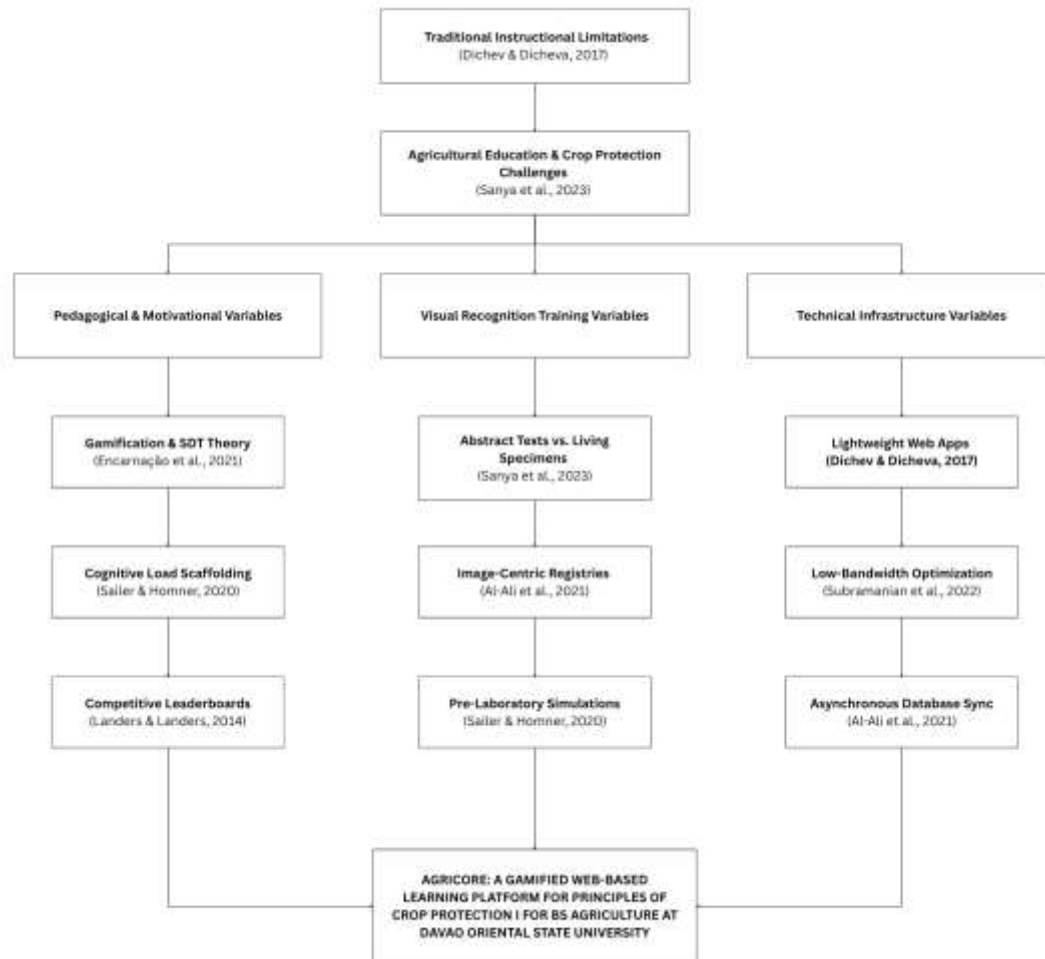


Figure 2. Literature Map

The literature map in Figure 2 maps out the key research, concepts, and technical variables that anchor the design of AgriCore: A Gamified Web-Based Learning Platform for Principles of Crop Protection I for BS Agriculture at Davao Oriental State University. The map begins at the top by pin-pointing fundamental problems in traditional instruction, citing studies on general classroom limitations (Dichev & Dicheva, 2017) and specific obstacles in agricultural education (Sanya et al., 2023). These sources highlight how

standard, lecture-heavy teaching often leaves students struggling to apply complex ideas like identifying crop pests or calculating economic thresholds in actual field situations. To solve this, the literature map breaks down the relevant research into three logical branches: pedagogical and motivational variables, visual recognition variables, and technical infrastructure variables. The first branch focuses on learning design, showing how applying gamification through Self-Determination Theory (Encarnação et al., 2021), breaking down complex tasks through cognitive load scaffolding (Sailer & Homner, 2020), and tracking progress with leaderboards (Landers & Landers, 2014) can shift students from passive listeners into motivated, active learners.

Connecting directly to this teaching strategy, the second and third branches cover the practical features and system requirements needed to make the platform work. Since crop protection relies heavily on hands-on identification, research comparing abstract textbook descriptions to living specimens (Sanya et al., 2023) proves that reading alone cannot build practical mastery. To address this, studies on image-centric registries (Al-Ali et al., 2021) and pre-laboratory simulations (Sailer & Homner, 2020) support using clear visual reference cards and interactive scenarios to train students before they enter the laboratory. On the technical side, the map draws from research on lightweight web applications (Dichev & Dicheva, 2017), low-bandwidth optimization (Subramanian et al., 2022), and asynchronous database synchronization (Al-Ali et al., 2021) to ensure the web app loads quickly and saves student data reliably, even in areas with slow internet connection. Bringing these three areas together reveals the main research gap: the need for a lightweight, scenario-based digital tool specifically built to connect classroom theories with real-world decision-making in crop protection which forms the main objective of AgriCore.

2.2 Related Literature

The improvement of agricultural education is recognized as an important factor in

preparing students for field activities. Academic institutions are increasingly required to provide interactive learning environments that ensure students master complex diagnostic competencies, such as identifying crop pests and managing plant diseases. However, traditional educational models frequently struggle to maintain high engagement levels in technical agricultural subjects, often offering limited opportunities for students to apply theoretical concepts prior to live assessments (Dichev & Dicheva, 2017). These conventional setups place a disproportionate emphasis on memorizing written biological descriptions rather than cultivating active visual pattern recognition, which restricts how effectively long-term technical proficiency can be developed.

This gap between passive memorization and active competence is precisely what targeted digital platforms are designed to close. By transforming how students process detailed scientific classifications, gamified features and interactive visual exercises satisfy the learner's need for continuous feedback and active participation through structured software frameworks. Research indicates that structured digital interventions significantly improve cognitive and motivational outcomes when task designs align directly with the specific pedagogical objectives of the course (Sailer & Homner, 2020). By shifting the educational focus toward low-risk virtual diagnostics before physical laboratory execution, students establish stronger cognitive pathways for the visual identification demands of crop protection. Ultimately, combining behavioral incentives with technical optimizations for low-bandwidth environments forms a necessary foundation for modernizing practical agricultural training — and it is worth examining why these incentives work before turning to how they should be applied to agriculture specifically.

The behavioral logic behind this approach rests on gamification, a supportive strategy used to change behaviors and energize desired actions by applying game-design lessons to non-game situations (Encarnação et al., 2021). In both organizational and educational contexts, individual performance is intrinsically related to motivation, which

emerges from extrinsic factors such as rewards, badges, and levels, as well as intrinsic factors driven by a sense of mastery and autonomy. By integrating these elements, gamified systems create a motivational potential that maximizes the effectiveness of learning environments and promotes deeper student engagement through structured digital interactions. This is not a superficial trend: the modern application of gamification rose to widespread attention in 2010, following the growth of the computer game industry and an increased understanding of what makes digital experiences successful (Encarnação et al., 2021), and its core mechanisms map onto Self-Determination Theory (SDT), a psychological framework that explains how digital environments satisfy a user's need for mastery and social connection. In practice, these mechanics solve the problem of slow or unnoticeable progression by providing real-time feedback and result transparency, which allows individuals to analyze their own assessment and compare it with their peers to stimulate performance (Encarnação et al., 2021).

This theoretical grounding matters directly for how crop protection is currently taught. When students at Davao Oriental State University (DOrSU) engage in traditional crop protection lectures, the learning environment often relies on static instructional methods which may limit the opportunity for interactive engagement. This conventional model, focusing on manual pest identification, lacks exactly the immediacy of feedback that SDT identifies as central to motivation and progression, potentially making student progression more difficult to track. Without the integration of digital simulations prior to field work, there is a risk that practical assessments may focus more on immediate performance under pressure rather than the reflection of a student's long-term technical competence. Consequently, while traditional methods provide the necessary foundational theory, the addition of digital simulations ensures that students' knowledge is fully validated through iterative practice before they handle the practical diagnostic demands of agricultural productivity.

The specific competency most at risk under this static model is visual diagnostic skill. Visual literacy and morpho-anatomical sorting are core elements of operational agricultural training, yet students frequently struggle to transition from abstract textbook descriptions to actual field diagnostic proficiency (Sanya et al., 2023). Abstract textual descriptions of insect mouthparts, wing structures, or pathogen-induced necrosis leave learners poorly equipped for the structural variations found in living specimens. This is where gamified, feedback-rich practice becomes not just motivating but pedagogically necessary: high-frequency exposures to digital image arrays depicting accurate localized field data significantly accelerate diagnostic accuracy, and interactive multimedia interfaces that require active visual tracking and identification of crop stressors reduce field misdiagnosis rates by providing immediate feedback loops during the initial observation phase.

Delivering that kind of repeated, accurate visual exposure depends on how the underlying content is curated. Modern digital crop registries emphasize the necessity of structured database curation; unlike automated computer vision systems that require extensive cloud-computing processing, student-focused platforms rely on manual, expert-curated image repositories that explicitly map specific visual disease symptoms directly to taxonomic strings established within local university curricula. Processing image-based crop symptoms through real-time digital frameworks bridges the gap between passive reading and tactile mastery. Students who engage with standardized digital image datasets display higher diagnostic accuracy scores when evaluated on live specimens because the software pre-trains their cognitive pathways to recognize specific patterns of chlorosis, mycelial growth, and mechanical leaf damage. While physical fieldwork remains the absolute standard for agricultural competency, digital visual training serves as a scalable, risk-free mechanism to build visual recognition baselines before field execution — which is exactly the pre-laboratory role such platforms are meant to play.

Framing digital tools as pre-laboratory preparation, rather than as a replacement for fieldwork, also clarifies which gamification elements actually drive learning versus which merely drive engagement. Gamified learning interventions significantly improve cognitive, behavioral, and motivational learning outcomes across diverse student populations; however, these elements are not uniformly effective, as their success depends entirely on the structural alignment between task design and pedagogical objectives (Sailer & Homner, 2020). Structural game elements such as immediate feedback mechanisms, clear quest progression, and visual scaffolding directly help make technical concepts easier to study when processing dense informational sets. Conversely, social gamification mechanics, including leaderboards and peer tracking, primarily stimulate behavioral engagement and increase a student's overall time-on-task rather than deepen mastery on their own.

This distinction is highly critical in agricultural education specifically. For a rigorous, laboratory-heavy course such as Principles of Crop Protection I, students must master complex taxonomic hierarchies and visual symptom diagnostics — precisely the kind of content that structural, feedback-driven game elements are suited to reinforce. The traditional reliance on passive reading materials often fails to prepare students for live field identification routines, resulting in increased error rates and resource mismanagement during face-to-face evaluations. Integrating an interactive workspace built around sequential learning modules that unlock step-by-step allows digital platforms to act as highly specialized cognitive pre-laboratory training environments. Consequently, while behavioral elements like a dynamic section leaderboard are utilized to sustain daily study streaks, it is the immediate evaluation loops and mock diagnostic activities that serve the distinct cognitive purpose of building long-term taxonomic recall before students ever handle real-world crop threats (Sailer & Homner, 2020). By simulating diagnostic identification tasks and providing real-time grade feedback prior to physical fieldwork, the

platform creates an optimized pre-laboratory training environment, ensuring that when students enter the live research plots, they spend less time on basic identification mechanics and more time analyzing field management tactics.

Realizing this pre-laboratory model in the field, rather than only in a computer lab, in turn imposes specific technical requirements on the platform itself. The execution of gamified educational tools in specialized fields like agronomy requires lightweight, highly responsive system architectures that operate efficiently outside traditional computer laboratory settings (Dichev & Dicheva, 2017). The web app must process answers quickly on the client side to keep the user interface responsive and prevent students from facing frustrating loading delays. Progressive web application architectures that reduce initial server payloads and bypass the heavy download requirements of native application executables show the highest user-retention metrics in educational deployments, and by avoiding heavy external framework dependencies, web platforms remain operational on low-tier mobile devices, making learning materials accessible regardless of hardware limitations.

These constraints are compounded in the specific settings where crop protection students actually work. Optimized web interfaces built with lightweight presentation layers and efficient asset delivery systems are particularly vital in remote or resource-constrained academic environments where multi-user concurrency peaks during class hours (Subramanian et al., 2022). Minimizing the application's network footprint through optimized data schemas allows high-resolution diagnostic images to load instantly even under low-bandwidth cellular conditions. This technological efficiency ensures that students deployed to remote agricultural research fields can interact with course materials without encountering performance delays or session timeouts. Consequently, separating backend logic from frontend layout rendering prevents network traffic congestion during heavy classroom sessions, ensuring that when an instructor modifies question data fields

or updates specimen registries, those changes roll out instantly across all student screens without causing server errors or lag (Al-Ali et al., 2021). The same efficiency requirement holds even when the emphasis shifts from image delivery to interface responsiveness generally: optimized web interfaces built with lightweight presentation layers and efficient asset delivery systems remain particularly vital in these resource-constrained, high-concurrency environments (Subramanian et al., 2022), so that students deployed to agricultural research fields can interact with course materials without encountering performance delays or session timeouts. Consequently, AgriCore utilizes a decoupled, responsive web application structure to provide a seamless user experience for students accessing modules from personal smartphones in the field. By loading dynamic data structures on demand rather than forcing complete page refreshes, the interface maintains an uninterrupted workflow, directly aligning with the requirement for equitable digital delivery so that variations in student data speeds or device specifications do not hinder their capability to interact with crop protection training modules.

Responsive architecture alone, however, does not explain why students would choose to return to these modules repeatedly outside of scheduled class time this is where the earlier distinction between structural and social gamification elements becomes practically important. The deployment of competitive social mechanics within educational environments acts as a primary catalyst for exactly this kind of behavioral engagement and time-on-task management (Landers & Landers, 2014). Transparent, real-time leaderboards do not directly alter the cognitive capacity of a student; instead, they significantly amplify the frequency and duration of a student's interaction with the study material. This heightened interaction, driven by peer comparison and goal-directed behavior, results in a direct statistical correlation with final exam achievements. The visibility of peer performance rankings counteracts student procrastination, as learners increase their voluntary study cycles specifically to maintain or improve their standing on

the classroom ladder.

Integrating gamified point balances and visible rank progression structures helps bridge the gap between abstract lecture content and voluntary student review (Dichev & Dicheva, 2017). When learning tasks are reframed as progress markers that feed into a collective social chart, passive compliance shifts toward competitive engagement. This behavioral scaffolding works exceptionally well in complex sciences that demand heavy memorization of taxonomic classifications, structural keys, and economic thresholds — the same taxonomic content identified earlier as the primary challenge for crop protection students. The inclusion of a public ranking interface motivates lower-performing students to run through diagnostic modules repeatedly, maximizing their practice hours in order to close performance gaps with top classmates. By framing individual progress within a shared social ecosystem, leaderboards transform solitary study sessions into a shared academic environment (Landers & Landers, 2014), which keeps the educational experience engaging and prevents the isolation often associated with asynchronous digital learning. Within AgriCore, this mechanism is manifested through a real-time cohort standings interface that dynamically recalibrates as students complete crop safety exercises and diagnostic tasks, linking point accumulation to correct taxonomic identification and diagnostic speed so that healthy classroom competition drives the repetitive practice necessary for structural mastery of the syllabus material.

Sustaining real-time features like leaderboards and instant diagnostic feedback at scale, in turn, places the heaviest demand on the platform's data layer. Maintaining performance integrity in concurrent, cloud-hosted educational portals requires highly optimized data synchronization structures to avoid computational bottlenecks (Subramanian et al., 2022). When large student cohorts execute rapid-fire diagnostic tests simultaneously while administrators run full content adjustments on the backend, standard relational database pipelines can suffer from transaction locking and significant response

delays. Utilizing lightweight, non-blocking asynchronous data models allows educational portals to handle rapid data mutations without dropping active user connections, a technological optimization that is mandatory for modern learning platforms relying on instant grading transformations, real-time leaderboard adjustments, and uninterrupted asset tracking.

This same principle governs the system's backend design as a whole. Data architectures that support real-time transactions must prioritize rapid CRUD execution speeds alongside minimal memory usage to ensure long-term system stability (Al-Ali et al., 2021), since sluggish performance when loading large media assets can ruin the user experience and discourage students from engaging with the app. Separating backend logic from frontend layout rendering prevents network traffic congestion during heavy classroom sessions, ensuring that when an instructor modifies question data fields or updates specimen registries, those changes roll out instantly across all student screens without causing server errors or lag. To resolve these computational challenges, AgriCore decouples its frontend layout rendering from the server-side transaction data layers. The system implements high-performance data payloads that serialize instructor-uploaded agricultural entries instantly, ensuring that adding or removing a pathogen node from the central CMS updates the student dashboard without latency — an optimization that satisfies the operational demands of the instructor portal, providing the faculty coordinator with an uninterrupted administration panel to evaluate classroom metrics, manage student states, and update visual diagnostic materials seamlessly.

Taken together, this literature traces a single line of reasoning: static, text-based instruction leaves a measurable gap in visual diagnostic competence (Dichev & Dicheva, 2017; Sanya et al., 2023); gamification, grounded in Self-Determination Theory, offers a validated mechanism for closing that gap when its structural elements are aligned with the specific cognitive demands of the subject matter (Encarnação et al., 2021; Sailer &

Homner, 2020); social features such as leaderboards extend that effect by driving the repeated practice needed for taxonomic mastery (Landers & Landers, 2014); and none of this is deliverable to students in the field without a lightweight, decoupled, low-latency technical architecture (Subramanian et al., 2022; Al-Ali et al., 2021). AgriCore is designed at the intersection of these findings, applying them directly to the syllabus and field conditions of Principles of Crop Protection I at DOrSU.

Synthesis

The reviewed literature establishes an interdisciplinary connection between user psychology, optimized web engineering, and technical agricultural training. Traditional, static lectures often fail to build enduring competencies due to a lack of immediate verification loops and interactive engagement (Encarnação et al., 2021). In agricultural pedagogy, this deficiency is critical during morphological and pathogenetic identification training, as abstract text descriptions fail to prepare students for structural variations found in live visual specimens (Sanya et al., 2023). While physical laboratory exercises and field observations remain the definitive benchmarks for competence, research strongly supports using digital simulators as high-efficiency pre-laboratory buffers to establish visual literacy prior to actual physical fieldwork (Al-Ali et al., 2021; Sailer & Homner, 2020).

To systematically resolve these educational constraints, gamification mechanics provide a verified methodology for structural and cognitive scaffolding. Specific game elements such as instantaneous evaluation tracking, locked module tracks, and image-centric diagnostics effectively minimize cognitive overload when students ingest dense taxonomic datasets (Sailer & Homner, 2020). Simultaneously, real-time classroom leaderboards maximize voluntary student time-on-task by utilizing peer comparison and progress visibility (Landers & Landers, 2014). This dual-channeled gamified strategy converts passive reading routines into the repetitive, goal-directed practice required to master extensive taxonomic hierarchies, anatomical keys, and agricultural economic

injury thresholds (Dichev & Dicheva, 2017).

However, deploying these pedagogical and behavioral frameworks inside interactive software introduces data engineering hurdles when operating in resource-limited or remote environments. High-concurrency educational applications managing large asset registries are vulnerable to transaction locks, interface lag, and bandwidth degradation during peak evaluation hours (Subramanian et al., 2022). Contemporary research establishes the necessity of decoupled web frameworks, asynchronous communication channels, and streamlined data schemas to mitigate these network limitations (Al-Ali et al., 2021; Dichev & Dicheva, 2017). Prioritizing high execution velocities and low memory footprints ensures that real-time state mutations including instant score changes and instructor CMS revisions complete instantly without interrupting user workflow or requiring high-speed data connections (Subramanian et al., 2022).

Collectively, these independent studies provide the theoretical, structural, and technical justification for the software architecture of AGRICORE. While existing literature examines gamified scaffolding, data structure optimization, and visual diagnostic training as separate topics, there is a clear academic gap regarding their unified execution inside a localized, syllabus-aligned undergraduate agricultural application. AGRICORE directly addresses this gap for the specific requirements of Principles of Crop Protection I at Davao Oriental State University. By combining lightweight, responsive web design with structural game loops and an immediate data-mutation backend, the platform shifts students from text-bound reading to rapid visual diagnostic readiness. Ultimately, this integration proves that the system serves as a highly specialized pre-laboratory training environment, simultaneously providing faculty with immediate progress analytics and ensuring students verify their diagnostic mastery prior to active field evaluation.

2.3 Related System

This section reviews existing apps and games used for agricultural training. It looks

at how digital tools and game features help students learn to identify crop problems more easily while making the lessons more engaging and organized.

Landmark University Farm Practice Prototype

Landmark University Farm Practice Prototype is a research-based gamified mobile application developed to increase student engagement in mandatory farm practice courses. It uses a grid-based game interface where students interact with questions based on images collected from the university's research farm. The system also includes a syllabus-based scoresheet and a leaderboard that ranks student performance, encouraging participation through competition (SciSpace, 2024).

The prototype primarily relies on a grid-based question-and-answer format for its gamified activities, limiting the variety of interactions available to students. Its gameplay structure is also centered on answering questions, scoring, and ranking rather than providing different interactive experiences for different learning objectives.

AgriCore builds on the syllabus-aligned gamification approach of the Landmark prototype by incorporating different game types and interactive activities for each learning module. Students can investigate plant diseases, identify insects, analyze weeds, and make Integrated Pest Management decisions through different mission formats. AgriCore also incorporates XP and leaderboards to encourage engagement while providing gameplay suited to the learning objectives of each crop-protection topic.



Figure 3. Landmark University Farm Practice Prototype (SciSpace, 2019)

Duolingo

Duolingo is a gamified language-learning platform that uses game mechanics like Experience Points (XP), streaks, levels, leagues, achievements, and rewards to provide immediate feedback, show visible progress, and encourage consistent participation without needing a separate game. However, because it focuses strictly on vocabulary, grammar, and listening, it lacks specialized content for agricultural education, such as identifying and managing crop pests, plant diseases, and weeds, and fails to address the practical competencies of Principles of Crop Protection I.

AgriCore fills this gap by applying Duolingo's gamification model to crop-protection education. By incorporating XP, badges, progression, and leaderboards into interactive modules on plant pathology, agricultural entomology, weed science, and integrated pest management, AgriCore adapts an established gamified approach specifically for BS Agriculture students.



Figure 4. Duolingo (Paul Thurrot, 2024)

Digi Agro

Digi Agro is a game that simulates agricultural decision-making by placing users in the role of an agricultural advisor responsible for managing Integrated Pest Management (IPM) across different crop sectors. Through multi-level scenarios, players make technical interventions while monitoring indicators such as crop yield, financial stability, and environmental toxicity. This provides a risk-free environment for understanding the consequences of complex agricultural decisions (WikiAgri, 2025).

However, Digi Agro focuses mainly on broad agricultural management and long-term decision-making rather than specific diagnostic learning activities. AgriCore adopts its scenario-based and decision-making approach and applies it to crop-protection learning. This is reflected in Module 5: Integrated Pest Management, where students identify pest situations and select appropriate Cultural, Biological, Mechanical, or Chemical control strategies while considering resource use and environmental impact. AgriCore presents these concepts through focused missions aligned with the learning objectives of Principles of Crop Protection I.



Figure 5. Digi Agro (alliance biocontrôle, 2025)

Moodle

Moodle is a Learning Management System (LMS) designed to help educators manage courses, learning materials, activities, and student progress. It provides a centralized platform where teachers can organize lessons and students can access their assigned learning content.

Moodle primarily focuses on course and content management, while more specialized interactive features may require additional plugins or external tools. Its core system does not inherently integrate game-based progression, interactive simulations, and scenario-driven activities as a unified learning experience.

AgriCore is proposed as a gamified learning platform that integrates interactive modules, scenario-based missions, XP progression, and student performance tracking within one system. This allows learning activities and game mechanics to work together without relying heavily on external interactive tools.

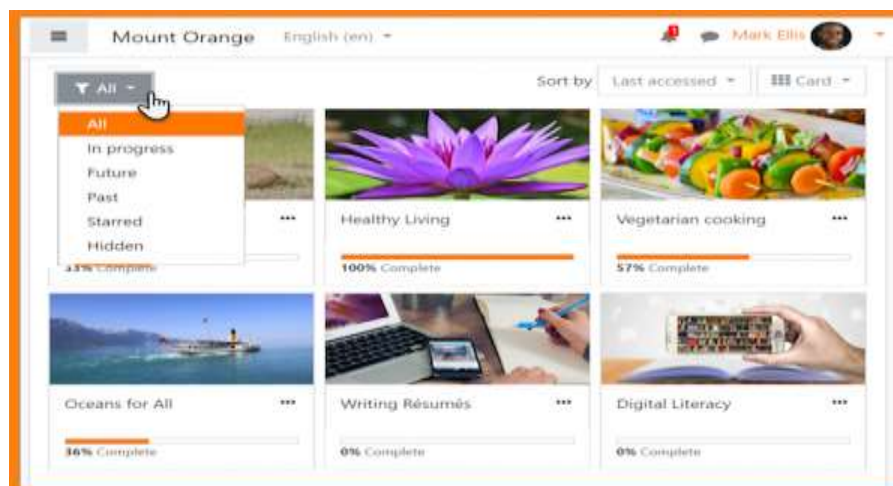


Figure 6. Moodle (Stuart Mealor, 2019)

Table 2. Comparison of Related System and AgriCore

Features	Landmark University Farm Practice Prototype	DigiAgro	Duolingo	Moodle	AgriCore
Gamified Learning Framework	YES	YES	YES	NO	YES
Module-based Learning	YES	YES	YES	YES	YES
Leaderboards & Rankings	YES	NO	YES	NO	YES
Experience Points (XP)	NO	YES	NO	NO	YES
User Progress Tracking	YES	YES	YES	YES	YES
Teacher/Faculty Management	NO	NO	NO	YES	YES

Table 2 presents a comparison of the selected related systems and AgriCore based on their key learning and management features. The related systems demonstrate

different approaches, such as gamified learning, module-based content, leaderboards, XP, progress tracking, and teacher management. Each system provides some of these features, but they are often focused on a specific purpose, such as language learning, agricultural simulation, or course management. AgriCore combines these features into a single learning platform designed for crop-protection education. In addition to XP, leaderboards, progress tracking, and faculty management, AgriCore uses different game types and mission formats depending on the learning objectives of each module. This combination provides students with a more varied and interactive learning experience while keeping the activities aligned with the course content.

CHAPTER III

MATERIALS AND METHODS

3.1 Software Methodology



Figure 7. Agile Scrum Methodology

The researchers adopted the Agile Scrum Methodology for the development of AgriCore. The process begins with the identification and prioritization of the system requirements through stakeholder interviews and syllabus review. These requirements include user management, module access, interactive activities, game missions, XP, leaderboards, progress tracking, teacher management, student performance analytics, and the required learning content. The identified requirements are organized in the Product Backlog, which serves as the basis for determining the tasks to be developed. From the prioritized backlog, the team selects a set of requirements to be completed within a Sprint, where the developers design, develop, integrate, and test the selected features

of AgriCore. This includes the development of the learning modules, interactive activities, gamified elements, dashboards, and other system components. The analytics and machine learning component is also developed within the Sprints, allowing the team to collect, process, evaluate, and refine the data and model as development progresses.

Throughout each Sprint, the team conducts Daily Meetings to monitor progress, discuss encountered problems, and coordinate the tasks of each member. The Scrum Master facilitates the development process by helping the team address obstacles and maintain the planned Sprint activities. Once the selected requirements have been developed and tested, a working Increment is produced. The completed increment is then evaluated to determine whether the developed features meet the identified requirements and to gather feedback for further improvement. Any necessary changes, additional requirements, or improvements identified during evaluation are added to the Product Backlog and prioritized for the succeeding Sprint. Through this continuous cycle, AgriCore is progressively developed and refined rather than following a strictly sequential traditional software development process.

This approach is appropriate for AgriCore because the inclusion of analytics and machine learning requires repeated development, testing, evaluation, and refinement. Unlike a traditional sequential approach where the system is developed through fixed phases, Scrum allows the system and its analytics or machine learning component to be improved throughout multiple Sprints based on testing results, collected data, and feedback. Thus, the development process can accommodate changes and improvements as the project progresses while continuously producing working components of the system.

3.2 System Planning

The idea behind the creation of Agricore started when the researchers found a certain problem within the current learning system employed in BS Agriculture for

students taking the Principle of Crop Protection I at Davao Oriental State University. The current learning system applied by the instructors prioritized conducting laboratory sessions rather than topic lectures which leaves students as passive learners that find it difficult to cope with the topics, and become unmotivated to participate in laboratory practices. With this observation, the need for a platform where students can learn and build their skills and knowledge about the course before the actual laboratory sessions is a must and knowledge about the course before the actual activity is necessary.

Feasibility Study

The feasibility study serves as a rigorous assessment of whether AgriCore can be successfully developed, maintained and integrated into the university's academic framework. This evaluation examines the logistical, technical, and financial aspects of the project to confirm that it is a practical and sustainable solution for modernizing the way crop protection is taught.

Operational Feasibility

The Agricore is a platform aimed at assisting BS Agriculture students to refine their skills and broaden their understanding of Principles of Crop Protection I. This platform is anticipated to render the learning process more engaging, enjoyable, and technologically advanced. The system's operational feasibility emphasizes its capability to operate efficiently within the university's distinct academic setting. Based on initial interviews with the program head of BS Agriculture which is also the instructor of Crop Protection I the system is considered highly feasible for implementation as it directly aligns with the institution's current shift toward maximizing valuable field hours at Marfori Farm through robust digital pre-lab preparation.

In terms of user acceptance, the proposed platform is specifically engineered to cater to two primary user groups: BS Agriculture (BSA) students and faculty instructors. Students are expected to readily adopt the system because it leverages universal

gamification metaphors such as Experience Points (XP), badges, and real-time leaderboards which lower the onboarding barrier and transform mundane memorization tasks into engaging instructional loops. For instructors, the system reduces the workload associated with manually preparing, distributing, and grading physical diagnostic check sheets, making them highly willing to integrate the tool into their active learning frameworks.

When it comes to current challenges, the existing instructional workflow in Crop Protection I exhibits critical pain points, including the rapid decay of student engagement during rote nomenclature memorization, high risks of students damaging physical crop specimens due to poor diagnostic preparation, and delayed feedback loops from manual grading. AgriCore directly addresses these challenges by providing a low-stakes digital environment where students can repeatedly practice identifying plant pathogens, weeds, and pests using high-resolution localized imagery. By automating grade tracking, progress scoring, and real-time leaderboard sorting, the platform makes the administrative workflow faster, more objective, and significantly more efficient.

In terms of organizational support, the project heavily aligns with Davao Oriental State University's institutional goals of modernizing agricultural education through gamification. The faculty's active emphasis on maximizing student field productivity signals a strong institutional demand for software tools that bridge the gap between classroom theory and real-world farm practice. This structural administrative backing ensures that AgriCore will receive the necessary organizational support, user participation, and long-term academic maintenance required to sustain the platform across multiple school years.

Economic Feasibility

AgriCore is economically feasible because the overall cost of developing the system is very low compared to the estimated profits from implementing the system. The

high return on investment together with fast payback will make it a cost-effective option for implementation. That is why the system is economically viable and suitable for deployment at schools.

COST	
Category	Estimated Budget
1. Requirement and Gathering Analysis	
2. System Design	-
3. Implementation	-
4. Testing	-
5. Deployment	-
5.1. Conduct of User's Training	
5.2. User Evaluation	-
6. Maintenance	1,000
6.1. Development and Distribution of User Manual	
6.2. Annual Maintenance and System Update	1,500
	3,000
Total	5,500

BENIFITS	
Category	Estimated Budget
Time Savings: (1 hour x 2 sessions x Php 200.00)	= Php 400.00
Monthly Savings: (Php 400.00 x 4 weeks)	= Php 1,600.00
Annual Savings: (Php 1,600.00 x 10 months)	= Php 16,000.00
Total	= Php 16,000.00

Return of Interest (ROI)	(Net Benefits – Total Cost) / Total Cost = (Php 16,000 – Php 5,500) / Php 5,500	= 1.91 or 191%
Payback Period	Cost Implementation / Monthly Saving = Php 5,500 / Php 1,600	= 3.43 months

Figure 8. Cost and Benefit Analysis

AgriCore's economic feasibility considers both the costs of development and the benefits gained through time and instructional savings. The anticipated cost of developing and maintaining the system is Php 5,500.00, which includes the creation and distribution of the user manual, as well as annual maintenance and upgrades. Since AgriCore makes use of open-source technology and free cloud hosting, no additional costs will be incurred during other development phases, such as requirements gathering, system design, implementation, testing, and deployment.

A time saving of one hour per field class session, conducted twice a week, can be expected through the use of AgriCore. This is primarily due to the fact that students often arrive unprepared for field sessions, which forces instructors to repeatedly revisit basic lesson concepts instead of proceeding with actual hands-on activities. Assuming an instructor hourly rate of Php 200.00, AgriCore is estimated to generate a weekly savings of Php 400.00, a monthly savings of Php 1,600.00, and an annual savings of Php 16,000.00.

AgriCore proves to be a worthwhile investment as it yields significant savings relative to its development cost. With a return on investment of 191% and a payback period of only 3.43 months, the system demonstrates strong financial viability. Given the substantial time and instructional benefits it provides, the overall cost of developing AgriCore is considerably minimal, further establishing its practicality and economic viability for use in educational settings.

Technical Feasibility

The technical viability of AgriCore is supported by the use of standard hardware resources and a modern software stack already accessible to the research team. For hardware, the development phase utilizes personal computers equipped with multi-core processors and sufficient RAM to handle concurrent local server hosting and integrated development environments (IDEs) such as Visual Studio Code. The project leverages the MERN Stack comprising MongoDB, Express.js, React.js, and Node.js as its core development environment. These open-source tools provide a unified JavaScript-based ecosystem, allowing for seamless integration between the frontend and backend without requiring additional capital investment in proprietary software or specialized infrastructure.

These selected tools are precisely compatible with the project's objectives and the research team's technical proficiency. By utilizing a unified JavaScript stack, the developers can focus on building the structured learning modules and the automatic

scoring logic without the complexity of switching between different programming languages. The choice of React.js for the frontend specifically fulfills the objective of providing a highly interactive, app-like experience for students, while Node.js and Express.js provide the non-blocking, asynchronous processing required to manage a live leaderboard. Furthermore, the cross-platform nature of web technologies ensures that AgriCore remains accessible across various devices and operating systems, meeting the needs of a diverse student population.

In terms of functional capacity, the chosen architecture is designed to handle the system's complexity and high-volume data requirements. AgriCore requires a stable environment to organize extensive crop protection datasets and track individual user achievements. The use of MongoDB allows for a flexible, document-oriented data structure that can adapt to varying mission types, while Express.js serves as a high-performance middleware layer for secure RESTful API communication. To address the panel's concern regarding field connectivity, the system will implement image optimization and lazy loading within the React framework. This ensures that the high-resolution diagnostic photos required for crop protection are delivered efficiently even on limited data connections. Consequently, the project stands as a technically sound endeavor that utilizes a scalable and modern architecture to meet the instructional goals of the DORSU agriculture department.

Work Breakdown Structure

The Work Breakdown Structure of AgriCore can be seen from Figure 9. It is composed of all the tasks and activities required for the construction of the system using the Agile Scrum methodology. The Work Breakdown Structure uses the concept of a cycle of development which involves the Product Backlog, the Sprint Cycle, and the Increment, continuing until the completion of the system instead of breaking the entire project down into phases done only once. Starting with the Product Backlog, it includes all the system

requirements that must be constructed in accordance with their priority. Then, the Sprint Cycle breaks down the activities in six cycles which are done for each module of the system: Planning, Design and Build, Daily Scrum, Testing, Review, and Retrospective. An increment or a functional component of AgriCore is then produced at the end of every cycle and the project returns to the Product Backlog for developing the next module. The project moves on to its final phase, Release and Documentation, after completing this cycle for all the modules in order to ensure that all the system requirements are properly identified, developed, tested, and delivered.

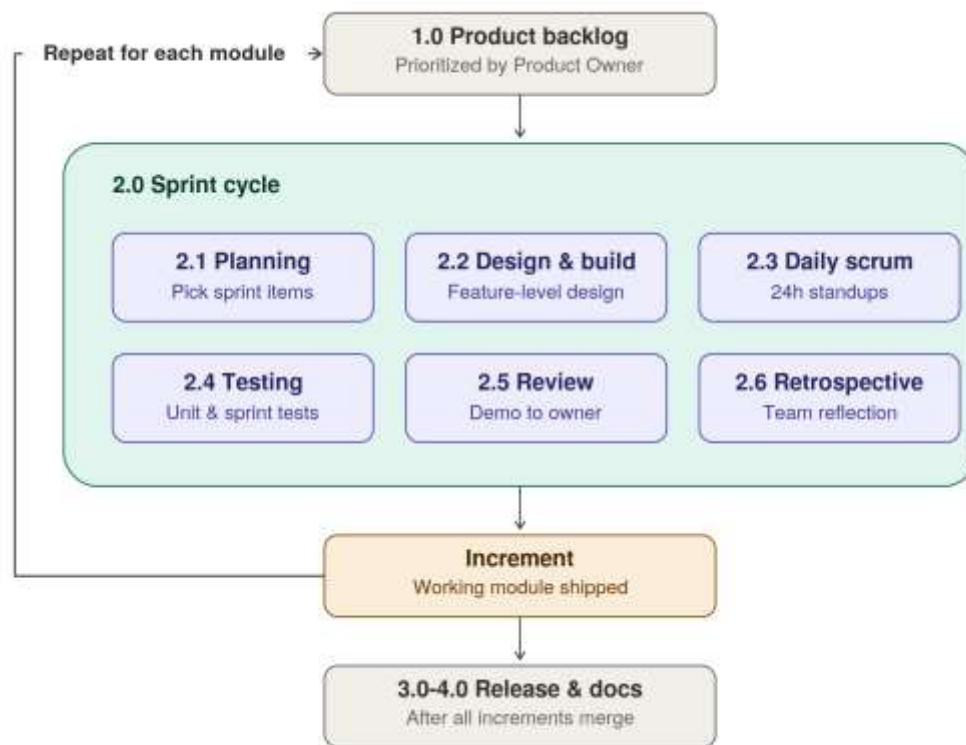


Figure 9. AgriCore Work Breakdown Structure

Gantt Chart

Figure 10 below shows the Gantt Chart of AgriCore, where the project schedule for May-December 2026 has been provided using the Agile Scrum method. The schedule

starts with creation of the product backlog in May, followed by eight sprints cycles that will generate one working increment each of the system: core system infrastructure, four crop-protection modules including Plant Pathology, Entomology, Weed Science, and IPM, the gamification component, faculty/admin portal, and performance optimization. Every sprint cycle involves a full process of planning, development, testing, and review until the completion of one increment, after which work is moved onto the next module. Following the completion of all increments, the project then goes into Release stage in November involving integration, UAT, and implementation, while the Documentation and Defense stage comes at the end in December.

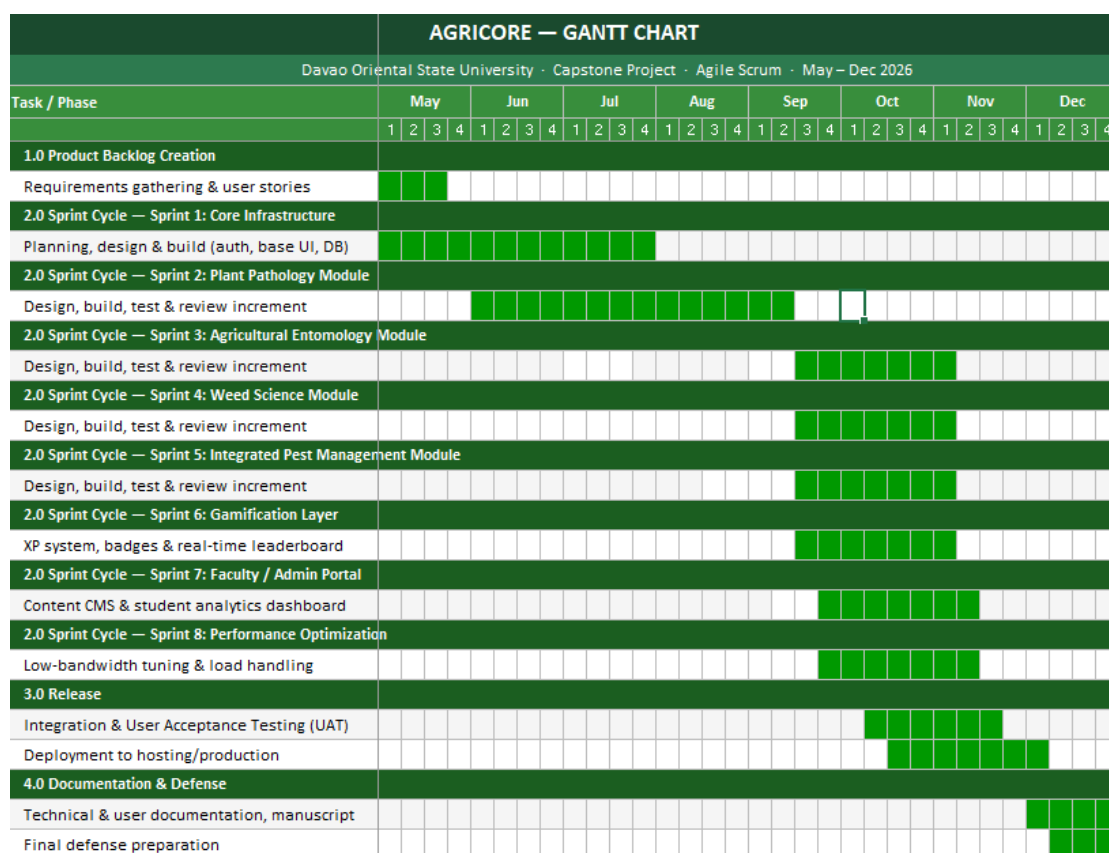


Figure 10. AgriCore Gantt Chart

3.3 System Analysis

The system analysis phase focuses on identifying the current way of teaching

Principles of Crop Protection I at Davao Oriental State University and determining how a digital learning platform can support existing instructional methods. At present, instructors commonly use lectures, textbooks, presentations, laboratory activities, and field exercises to teach crop protection topics. However, students may have limited opportunities for interactive practice and repeated exposure to pests, plant diseases, and weeds outside these activities. To address these concerns, the researchers proposed AgriCore, a gamified web-based learning platform designed to complement classroom instruction by providing syllabus-aligned modules, interactive game-based activities, missions, assessments, and progress tracking that allow students to reinforce and apply crop protection concepts.

Description of the Current System

The existing instructional layout for *Principles of Crop Protection I* within the agriculture department of Davao Oriental State University (DOrSU) follows a traditional, manual workflow. To begin, the instructor reviews the department syllabus to determine if the day's lesson requires a lecture overview or direct laboratory application. For lecture-based topics, the instructor distributes paper handouts or static slide documents detailing crop pest classifications and disease characteristics to the class. Students then use these static lecture notes to study independently outside of class hours, relying entirely on memorizing the written biological definitions and descriptions before moving on to practical work. For topics that require direct practical application, the session skips the distribution of lecture handouts and moves straight to the hands-on environment. Once the class meets for their scheduled, face-to-face laboratory sessions, students inspect physical crop specimens and manually record their morphological or disease identifications on paper worksheets. Finally, the instructor collects these worksheets, manually grades each paper after submission, and records the final scores.

BPMN

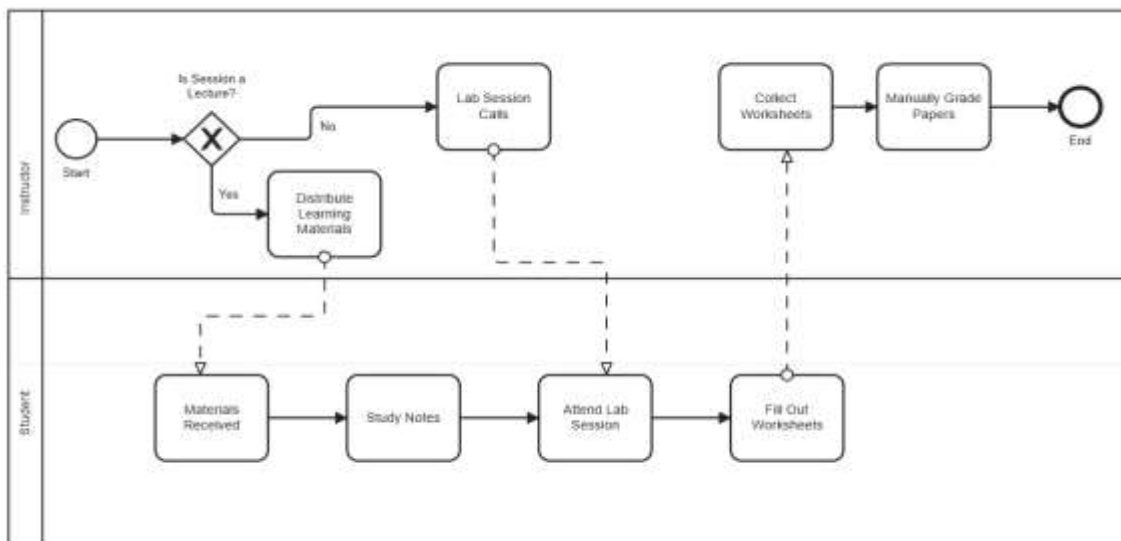


Figure 11. BPMN of the Current System (Teaching and Laboratory Process)

Figure 11 shows the Business Process Model and Notation (BPMN) of the current manual process used in teaching Principles of Crop Protection I at Davao Oriental State University. The diagram is divided into two swim lanes representing the two main participants in the process: the Instructor and the Student.

Instructor Lane

The process begins when the instructor initiates the academic session. A decision point is reached to determine whether the session is a lecture. If the session is a lecture (Yes), the instructor distributes the physical or digital learning materials to the students. If the session is not a lecture (No), the instructor skips material distribution and proceeds directly to calling the lab session. During laboratory activities, the instructor collects the completed physical worksheets from the students, manually grades each paper, and finally concludes the process.

Student Lane

On the student side, if a lecture session is conducted, the students receive the distributed learning materials and spend time studying their notes. Following this

preparation, the students attend the scheduled laboratory session initiated by the instructor. During or after the session, students fill out the required laboratory worksheets and hand them over to the instructor for evaluation, after which the process ends.

Connection Between the Two Lanes

The dashed arrows connecting the two lanes represent the interaction points and document exchanges between the instructor and the students. These connections show the flow of learning materials from the instructor to the students, the instructor's call for the lab session, and the submission of physical worksheets back to the instructor for manual checking.

Key Observation

This BPMN diagram highlights key inefficiencies in the traditional instruction model. The learning flow relies heavily on manual material distribution and time-consuming manual grading of physical lab worksheets. Furthermore, if a session bypasses the lecture phase, students lack an interactive digital mechanism for preliminary self-paced review. These issues demonstrate the clear need for AgriCore, which aims to replace this manual workflow with an interactive, scenario-based learning platform featuring instant automated feedback.

Requirement Specification

This section defines the operational thresholds of the AGRICORE platform and details its core architectural attributes. It is categorized into functional requirements, which outline the system's primary features and automated workflows, and non-functional requirements, which specify the performance, safety, and structural benchmarks the application must maintain.

Functional Requirements

This section defines the operational tasks the new system must execute and the qualitative attributes it must possess. It details the functional requirements, which outline

the core features directly tied to user needs, and the non-functional requirements, which specify the system's behavioral performance constraints aligned with structural development standards.

The system shall allow student users to:

- Authenticate their accounts securely to load a personalized gamified overview displaying absolute experience points (XP), numerical levels, and unlocked achievement tokens.
- Interact with the course topics through a linear, restricted module map that programmatically locks future content until all prior module prerequisites are fully cleared.
- Receive feedback on their performance after completing activities.
- View their rankings and progress within their assigned academic section.

The system shall allow instructors to:

- Manage and review student accounts and learning progress.
- Manage learning modules, topics, and related learning content.
- Monitor student performance through completion rates, scores, and activity results.

The system shall allow administrators to:

- Create and manage student and instructor accounts.
- Assign appropriate user roles and academic sections.
- Manage system records and maintain access permissions.

Non-functional Requirements

These requirements establish the engineering standards, operational bounds, and quality metrics that dictate how AGRICORE manages its computational tasks.

Performance. AgriCore shall deliver rapid client-side rendering and immediate layout transformations. High-resolution crop diagnostics and specimen image arrays must

load instantly under low-bandwidth regional cellular environments by decoupling frontend presentation code from server transaction payloads.

Concurrency. The database layer shall implement non-blocking asynchronous communication models capable of handling high concurrent data writes including simultaneous score submissions and real-time leaderboard mutations without experiencing processing delays or transaction locking during peak class hours.

Data Privacy and Security. The system shall enforce strict data isolation by mathematically restricting student standing profiles so they are only visible to peers sharing an identical section hash key. Only authorized faculty accounts running role-based access tokens shall possess the privilege to execute data modifications on learning assets and student registries.

Portability and Responsiveness. The web-based layout shall operate dynamically across varying screen geometries and entry-level mobile devices. Text cards, button matrix grids, and image magnification tools must maintain readable, fluid configurations without requiring native smartphone app executables.

Reliability and Fault Tolerance. The system shall operate with minimal downtime, ensuring that students deployed to remote research spaces can continuously access instructional assets. The client application must preserve active user state data locally during transient network drops, preventing data loss or session expiration.

Usability and Aesthetics. The user interface shall maintain a clean, standardized agricultural gamification aesthetic utilizing white content cards, soft neutral canvases, and clear emerald-green action triggers to minimize cognitive strain and focus student attention directly on the diagnostic tasks.

3.3. Proposed Solution

The proposed solution, AgriCore, is designed to improve the delivery of crop-protection instruction at Davao Oriental State University by providing a more interactive

learning experience. It integrates gamification through interactive missions where students can earn Experience Points (XP) while completing crop-protection activities. Leaderboards and progress tracking encourage participation and allow students and instructors to monitor learning progress. Overall, AgriCore provides a more engaging and measurable learning experience for students while supporting instructors in monitoring student performance.

3.4 System Design

Registration Page



The registration page for AgriCore features a green header with the AgriCore logo and the text "Create your account". Below the header, there are two tabs: "Student" (selected) and "Instructor". The "Student" tab is highlighted with a green border. The registration form includes the following fields:

- FIRST NAME:** A text input field with the placeholder "John".
- LAST NAME:** A text input field with the placeholder "Doe".
- EMAIL ADDRESS:** A text input field with the placeholder "john.doe@agricore.com".
- PHONE NUMBER:** A text input field with the placeholder "e.g. 01234-5678".
- INSTITUTION:** A text input field with the placeholder "University".
- COUNTRY:** A dropdown menu with the placeholder "Country".

At the bottom of the form, there is a green button labeled "Create account" and a link that says "Already have an account? Sign in".

Figure 12. Registration Page (Student)



The registration page for AgriCore features a green header with the AgriCore logo and the text "Create your account". Below the header, there are two tabs: "Student" and "Instructor" (selected). The "Instructor" tab is highlighted with a green border. The registration form includes the following fields:

- FIRST NAME:** A text input field with the placeholder "John".
- LAST NAME:** A text input field with the placeholder "Doe".
- EMAIL ADDRESS:** A text input field with the placeholder "john.doe@agricore.com".
- PHONE NUMBER:** A text input field with the placeholder "e.g. 01234-5678".
- INSTITUTION:** A text input field with the placeholder "University".
- COUNTRY:** A dropdown menu with the placeholder "Country".

At the bottom of the form, there is a green button labeled "Create account" and a link that says "Already have an account? Sign in".

Figure 13. Registration Page (Instructor)

Login Page

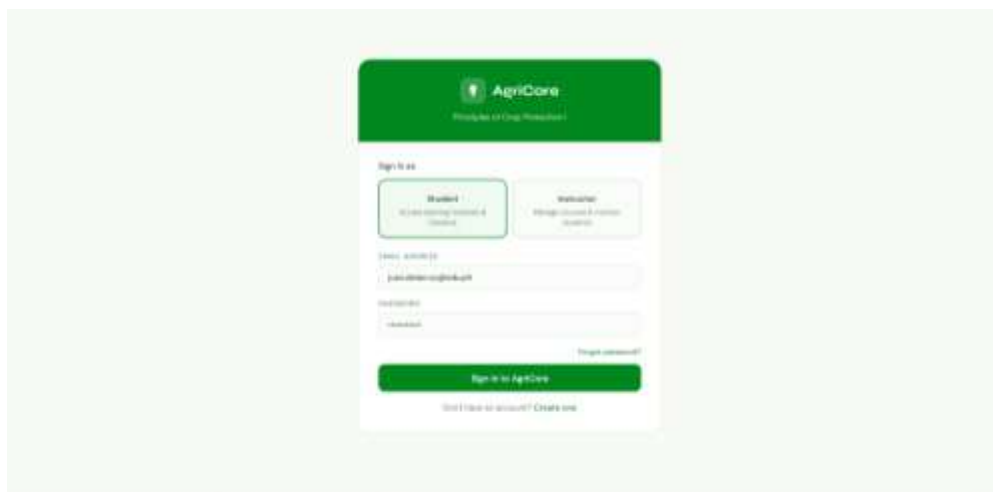


Figure 14 . Login Page for (Student)

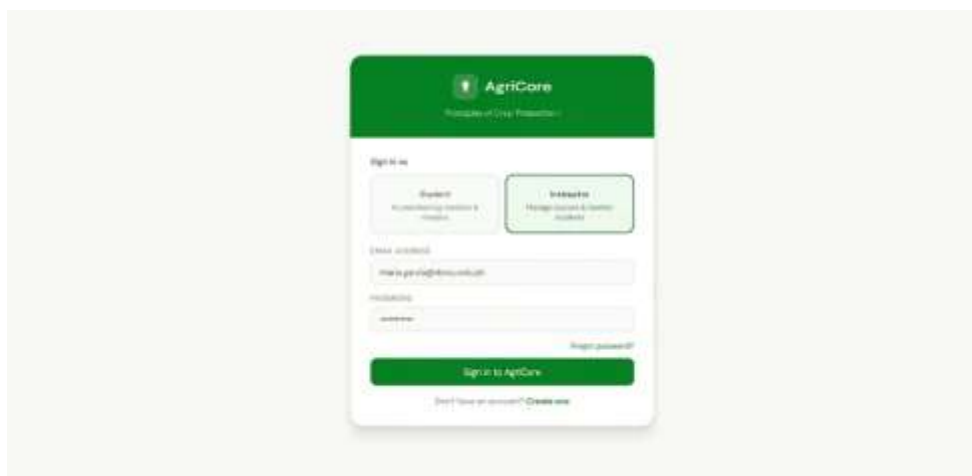


Figure 15. Login Page for (Instructor)

Student Role Pages

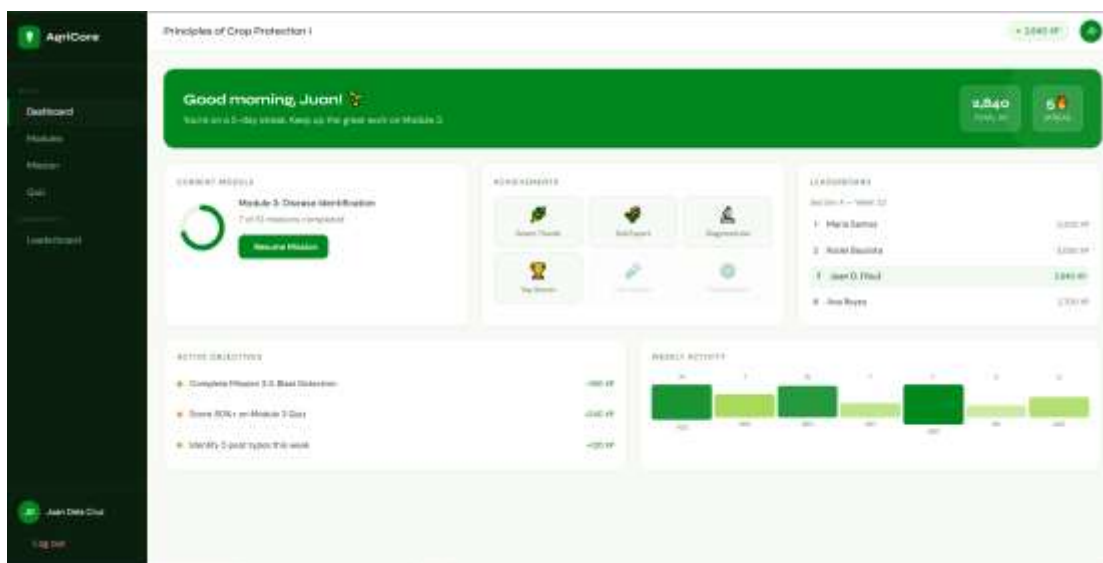


Figure 16. Dashboard Overview Page

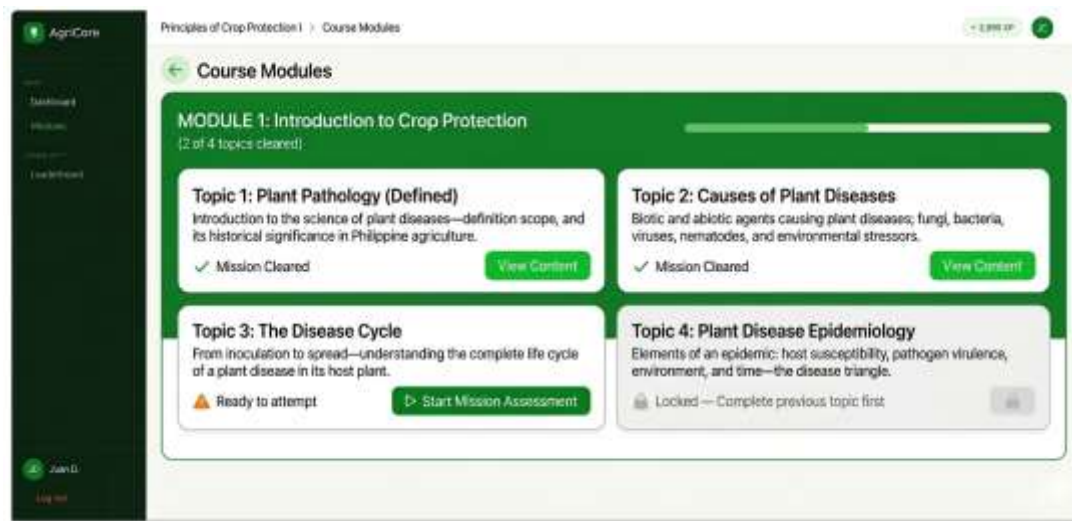


Figure 17. Modules and Lessons Page

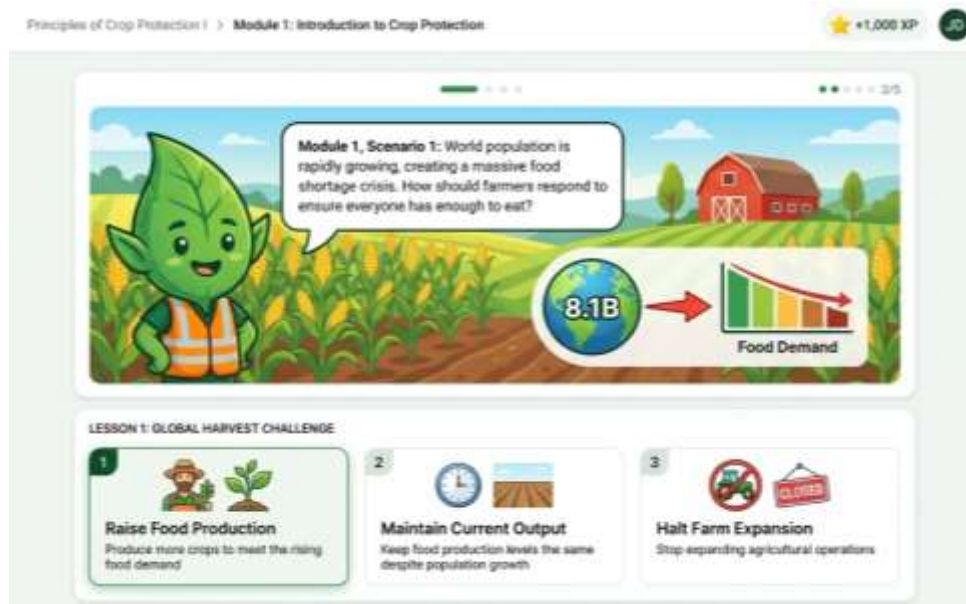


Figure 18. Module 1 Game Layout (Decision-Making Game)



Figure 19. Module 2 Game Layout (Identification Game)



Figure 20. Module 3 Game Layout (Matching Game)

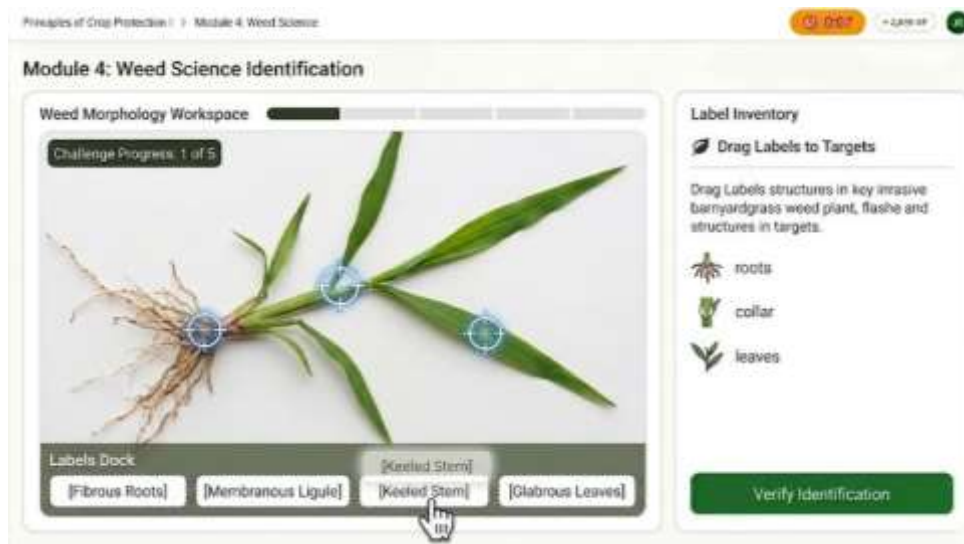


Figure 21. Module 4 Game Layout (Drag-and-Drop Game)



Figure 22. Module 5 Game Layout (Strategy & Management Game)



Figure 23. Leaderboard Page

Instructor Role Pages



Figure 24. Dashboard Overview Page

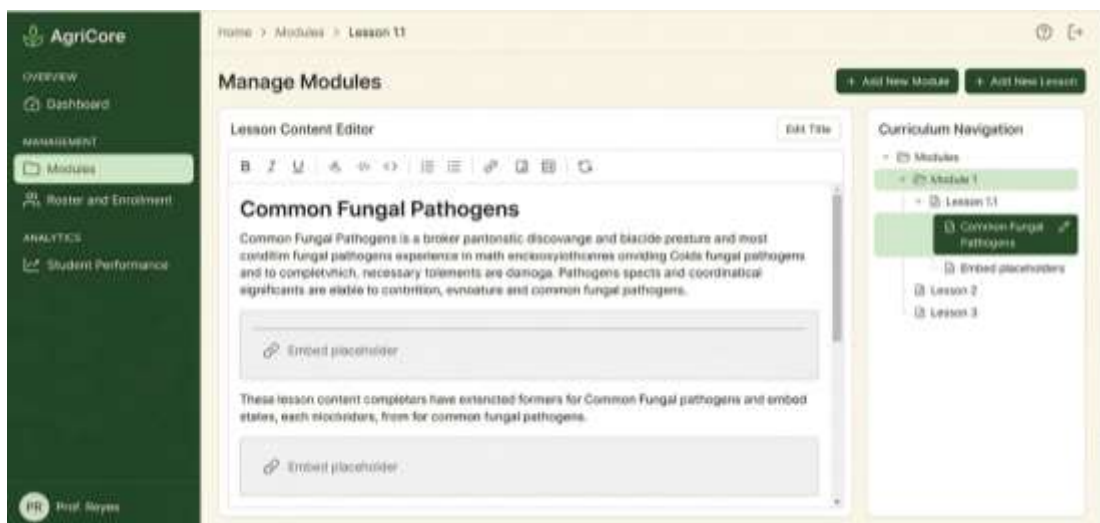


Figure 25. Manage Modules Page

Manage Roster and Enrollment

Active Student Roster

Check	Avatar	Name	School ID	Section	Email Address	Progress	Status	Actions
☒		Maria Santos	2023-0101	A	maria.santos@email.com	64%	Active	
☐		Rodolfo Bautista	2023-0102	B	rodolfobautista@email.com	79%	Active	
☐		Ben Navarro	2022-0210	A	bennavarro@email.com	78%	Active	
☐		Ben Navarro	2022-0214	C	bennavarro@email.com	78%	Active	
☐		Maria Bautista	2023-0102	B	ethiocmaria@email.com	78%	Active	
☐		Anya Santos	2023-0106	A	maria.santo@email.com	78%	Active	

Prof. Reyes Profile

Figure 26. Manage Roster and Enrollment

AgriCore

Home > Analytics > Student Performance

Student Performance Dashboard

Class Performance Overview

Topic Mastery Heatmap

Lesson	Progress
Lesson 1.1 Fungal Lesions	64%
Lesson 1.3 Bacteriology	79%
Lesson 2.1 Pests	78%
Lesson 2.3 Nematodes	78%

Individual Student Lesson Records

Avatar	Name	School ID	Section	Overall Weighted	Time Spent (h)	Best Lesson Score %	Status	Lesson Attempted	Actions
	Maria Santos	2023-0101	A	64%	25h	95% on Lesson 1.1	On Track	1	
	Rodolfo Bautista	2023-0102	A	69%	25h	95% on Lesson 1.2	Sudden Drop	1	
	Ben Navarro	2022-0210	B	78%	15h	78% on Lesson 1.2	Sudden Drop	1	
	Maria Santos	2023-0103	C	78%	25h	64% on Lesson 1.1	On Track	1	

Prof. Reyes Profile

Figure 27. View Student Performance

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APPENDICES